LARA: Layer-Aware Representation Alignment

One shared predictor across randomly sampled layer pairs

Diffusion Transformer

Transformer blocks

Target layer b

A_b

Source layer a

A_a



Noisy input x_t

01 Prediction

1 random pair (a, b) per training iteration

sg

Match the target

L_{pred}

$\hat{A}_{a \to b}$

Shared predictor Gψ

Time + layer emb.

Condition on timestep and layer depths

Random layer pairs

examples across iterations

Iteration 1: A_2 → A_7

Iteration 2: A_4 → A_{10}

Iteration 3: A_6 → A_9

...

LARA objectives

02 Transition consistency

a < b < c

Predictor-only update

Detached source: $sg(A_a)$

Online prediction: Online: a → c

EMA update

Cached prediction: $\hat{A}_{a \to b}$

sg

EMA: b → c

Target (sg)

L_{cons}

03 Redundancy

1 random pair (i, j) per training iteration

Sampled independently of (a, b)

A_i

A_j

L_{div}

sg: stop-gradient

EMA: moving-average predictor